

Sec. VI-E, revised: RMS residual bound of the PNLS fit

preview build; see the header comment for the numbering map

E. RMS residual bound of the PNLS fit

The envelope (90) bounds the deviation e_ω pointwise; this subsection converts it into a certificate on the quantity the pipeline actually reports, the residual RMS of the deployed fit. The argument needs one observation and one integral.

The witness inequality The record in the window is $y(\tau) = \omega_{\text{nom}}(\tau) + e_\omega(\tau) + b + n(\tau)$, with τ counted from the *realised* onset t_{on} — the instant the restoring moment reverses at the vehicle, wherever the actuation chain places it relative to the commanded ramp clock. The PNLS stage fits the family $\hat{h}(\tau; \theta) = C_1^* (\cosh(C_2^*(\tau - t^*)) - 1) \mathbf{1}[\tau \geq t^*] + C$ with $\theta = (C_1^*, C_2^*, t^*, C)$ free, minimising the objective of (92). The nominal solution is itself a member of this family: the choice $\theta_0 = (C_1, C_2, t_{\text{on}}, b)$ reproduces $\omega_{\text{nom}} + b$ exactly. A minimum never exceeds the value at one feasible point, so the fit cannot lose to this *witness*, and

$$\text{RMS}(r) \leq \text{RMS}(e_\omega + n) \leq \text{RMS}(e_\omega) + \text{RMS}(n). \quad (93)$$

No property of the minimiser is used — the fitted parameters may wander along the amplitude-onset ridge of Sec. V — and no admissibility condition arises, because the witness is the nominal itself rather than a shifted surrogate. Two consequences are worth stating. First, a pure actuation delay shifts t_{on} and nothing else; since t^* is free, the witness adopts the realised onset and the delay never enters the bound — no latency term is required. Second, before the onset both the witness and the record sit at the baseline, so the pre-onset segment contributes only noise: no pre-onset penalty of the form (108) appears, and extending the RMS to the full window only dilutes the left side of (93).

The window RMS of the envelope Squaring the envelope (90) and integrating over the window, with $x = C_2 \tau_{\text{end}}$,

$$\text{RMS}(e_\omega) \leq \bar{\rho} K C_2 \sqrt{\frac{B(x)}{x}}, \quad B(x) = \frac{\sinh 2x}{4} - \frac{x}{2}, \quad (94)$$

$B(x) = \int_0^x \sinh^2 u \, du$ by the double-angle identity; $\sqrt{B(x)/x}$ is the window RMS of the growing mode $\sinh(C_2 \tau)$, and x follows per ramp rate from the design sweep $\sinh x - x = \varphi_{\text{max}} W_{zCoM} C_2 / \dot{M}$ of Sec. VI-D. The tilt cap φ_{max} must dominate the tilt the windows actually reach, and is set per campaign: 5° in simulation, where the gate stops the window near 4° , and 8° on hardware, where the slightly uneven ground of the test site lets the in-window excursion reach 7.7° . Every appearance of φ_{max} is monotone, so a dominating cap only over-covers.

The measured decomposition and the onset shift For the critical-moment readout of Sec. VI-F the deviation is split against the growing mode,

$$e_\omega(\tau) = \beta \sinh(C_2 \tau) + \delta e_\omega(\tau), \quad (95)$$

with the end-matched coefficient

$$\beta = \frac{e_\omega(\tau_{\text{end}})}{\sinh(C_2 \tau_{\text{end}})}, \quad (96)$$

so that δe_ω vanishes at both window ends. [(95)–(97) are function-level identities in τ ; the sample index i belongs to (92) alone.] The envelope (90) caps the coefficient a priori,

$$|\beta| \leq \frac{\bar{\rho}}{J_P C_2}, \quad (97a)$$

and the sinh multiple is exactly the tangent direction of the onset shift, so the fitted branch absorbs it as a critical-time offset

$$\tanh(C_2 \delta t_c) = -\frac{\beta}{C_1}, \quad \delta t_c < 0, \quad (97b)$$

which Sec. VI-F converts into the critical-moment error $|\Delta M_{\text{crit}}| = \dot{M} |\delta t_c|$. This absorption is also why the measured residuals stay flat as the ramp rate grows: the dominant, exponentially growing part of e_ω is precisely the component the family's onset freedom removes.

The noise term What the family cannot represent and the dynamics do not produce remains in the residual as noise. On hardware the rotor-induced vibration has no pre-experiment model, so its amplitude is read from the run's own out-of-band content: with n_{hi} the residual RMS above $f_c = 5$ Hz — a band the cosh family cannot occupy, its fastest scale being $C_2/2\pi \approx 0.8$ Hz — the full-band estimate is

$$\text{RMS}(n) = \text{RMS}(n_{\text{hi}}) \sqrt{1 + (s_{\text{med}} \kappa_q)^2}, \quad (98)$$

with the quiet-segment band ratio and in-window enhancement of Sec. VI-D ($s_{\text{med}} \kappa_q = 1.31$ on the campaign). In simulation the sensor noise is declared, Gaussian with $\sigma_n = 2.450 \times 10^{-4}$ rad/s = $0.014^\circ/\text{s}$ — an order below the smallest envelope — so the noise term is dropped there.

The bound, and its validation Combining (93), (94) and (98),

$$\text{RMS}(r) \leq \bar{\rho} K C_2 \sqrt{\frac{B(x)}{x}} + \text{RMS}(n), \quad (99)$$

with every quantity on the right a pre-experiment design constant except the hardware n_{hi} . Figs. 15 and

16 test (99) on every run of both campaigns, the measured side being the deployed free fit exactly as it runs in the pipeline — both shape constants and the onset free, full window, no calibration on either side. In simulation ($\varphi_{\max} = 5^\circ$, envelope alone) 307 of 308 runs sit inside; the single exceedance is the case whose injected offset (38, 14) mm lies beyond the design rectangle assumed by $\bar{\rho}$, at the fastest ramp — the certificate fails exactly where its own premise does. On hardware ($\varphi_{\max} = 8^\circ$, envelope plus (98)) all 140 runs sit inside, the worst using 0.83 of its cap. The median used fraction is 0.32–0.33 on both campaigns: one construction, validated on both platforms, the panels differing exactly in the channels the platforms differ in.

Remark An earlier version of this analysis bounded δe_ω itself through the Dirichlet response of the forcing channels [the former (98)–(108)]. Measured directly, that cap bounds the in-family forcing response but not the realised remainder, whose out-of-family content is amplified by the same sinh growth the envelope charges for; it is therefore omitted, and the certificate on the residual is (99) alone. The decomposition (95)–(97b) retains its two roles: the a-priori ceiling on the onset shift consumed by Sec. VI-F, and the explanation of the residual’s rate-flatness.